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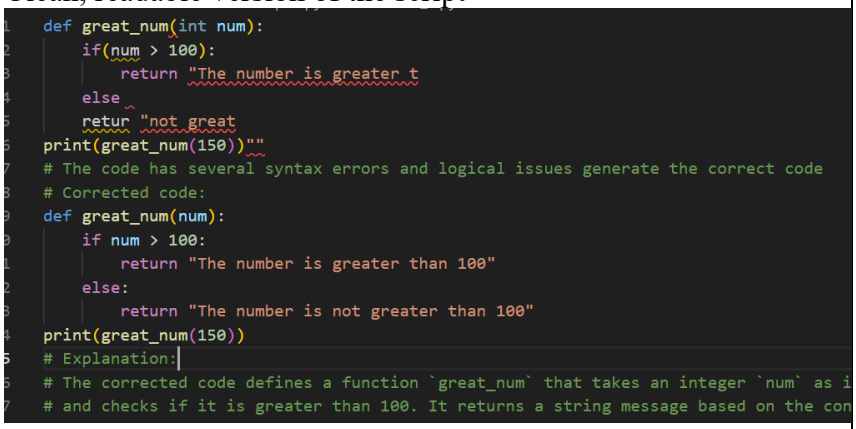
AssignmentNumber:10.4		
Q.No.	**Question**	***ExpectedTime to complete***
1	**Lab 9 – Code Review and Quality: Using AI to Improve Code Quality and Readability** **Lab Objectives** • Use AI for automated code review and quality enhancement. • Identify and fix syntax, logical, performance, and security issues in Python code. • Improve readability and maintainability through structured refactoring and comments. • Apply prompt engineering for targeted improvements. • Evaluate AI-generated suggestions against PEP 8 standards and software engineering best practices	Week 5
	Task 1: AI-Assisted Syntax and Code Quality Review **Scenario** You join a development team and are asked to review a junior developer's Python script that fails to run correctly due to basic coding mistakes. Before deployment, the code must be corrected and standardized. **Task Description** You are given a Python script containing: • Syntax errors • Indentation issues • Incorrect variable names • Faulty function calls Use an AI tool (GitHub Copilot / Cursor AI) to: • Identify all syntactic and structural errors • Correct them systematically • Generate an explanation of each fix made **Expected Outcome** • Fully corrected and executable Python code	

- AI-generated explanation describing:
  - Syntax fixes
  - Naming corrections
  - Structural improvements
- Clean, readable version of the script



```
def great_num(int num):
    if(num > 100):
        return "The number is greater t
    else
    retur "not great
print(great_num(150))""
# The code has several syntax errors and logical issues generate the correct code
# Corrected code:
def great_num(num):
    if num > 100:
        return "The number is greater than 100"
    else:
        return "The number is not greater than 100"
print(great_num(150))
# Explanation:
# The corrected code defines a function `great_num` that takes an integer `num` as i
# and checks if it is greater than 100. It returns a string message based on the con
```

Task 2: Performance-Oriented Code Review

Scenario

A data processing function works correctly but is inefficient and slows down the system when large datasets are used.

Task Description

You are provided with a function that identifies duplicate values in a list using inefficient nested loops.
Using AI-assisted code review:

- Analyze the logic for performance bottlenecks
- Refactor the code for better time complexity
- Preserve the correctness of the output

Ask the AI to explain:

- Why the original approach was inefficient
- How the optimized version improves performance

Expected Outcome

- Optimized duplicate-detection logic (e.g., using sets or hash-based structures)
- Improved time complexity
- AI explanation of performance improvement
- Clean, readable implementation

```python
def print_duplicates(numbers):
    duplicates = []
    for i in range(len(numbers)):
        for j in range(i + 1, len(numbers)):
            if numbers[i] == numbers[j] and numbers[i] not in duplicates:
                duplicates.append(numbers[i])
    return duplicates
numbers = [1, 2, 3, 4, 5, 2, 3, 6]
print("Duplicate numbers:", print_duplicates(numbers))
#Make this code more efficient by using a set to track seen numbers and duplicates.
def print_duplicates(numbers):
    seen = set()
    duplicates = set()
    for number in numbers:
        if number in seen:
            duplicates.add(number)
        else:
            seen.add(number)
    return list(duplicates)
numbers = [1, 2, 3, 4, 5, 2, 3, 6]
print("Duplicate numbers:", print_duplicates(numbers))
#Explaination:
#The original code uses a nested loop to compare each number with every other number,
#which results in a time complexity of O(n^2). The improved code uses a set to track see
#and duplicates, which allows it to check for duplicates in O(1) time.
```

```
OBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

\python\10_4.py'
uplicate numbers: [2, 3]
uplicate numbers: [2, 3]
```

Task 3: Readability and Maintainability Refactoring

Scenario

A working script exists in a project, but it is difficult to understand due to poor naming, formatting, and structure. The team wants it rewritten for long-term maintainability.

Task Description

You are given a poorly structured Python function with:

- Cryptic function names
- Poor indentation
- Unclear variable naming
- No documentation

Use AI-assisted review to:

- Refactor the code for clarity
- Apply PEP 8 formatting standards
- Improve naming conventions
- Add meaningful documentation

Expected Outcome

- Clean, well-structured code

- Descriptive function and variable names
- Proper indentation and formatting
- Docstrings explaining the function purpose
- AI explanation of readability improvements

```python
#Cryptic function names,Poor indentation,Unclear variable naming,No documentation
#Your task is to refactor the code to improve its readability and maintainability while preserving its
def calculate_area(radius):
    """Calculate the area of a circle given its radius."""
    import math
    area = math.pi * (radius ** 2)
    return area
def calculate_circumference(radius):
    """Calculate the circumference of a circle given its radius."""
    import math
    circumference = 2 * math.pi * radius
    return circumference
def main():
    """Main function to execute the area and circumference calculations."""
    radius = float(input("Enter the radius of the circle: "))
    area = calculate_area(radius)
    circumference = calculate_circumference(radius)

    print(f"The area of the circle is: {area:.2f}")
    print(f"The circumference of the circle is: {circumference:.2f}")
if __name__ == "__main__":    main()
#Explaination:
#The refactored code is now more readable and maintainable. The function names have been
#changed to be more descriptive of their purpose, and the variable names have been improved for clarity
#Additionally, docstrings have been added to each function to explain their functionality, and
#the main function has been structured to clearly show the flow of the program.
```

```
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se) PS C:\Users\mukes\OneDrive\Desktop\python>  c:; cd 'c:\Users\mukes\OneDrive\Desktop\python'; & 'C:\Users\mukes
er the radius of the circle: 4
 area of the circle is: 50.27
 circumference of the circle is: 25.13
se) PS C:\Users\mukes\OneDrive\Desktop\python>
```

Task 4: Secure Coding and Reliability Review

Scenario

A backend function retrieves user data from a database but has security vulnerabilities and poor error handling, making it unsafe for production deployment.

Task Description

You are given a Python script that:

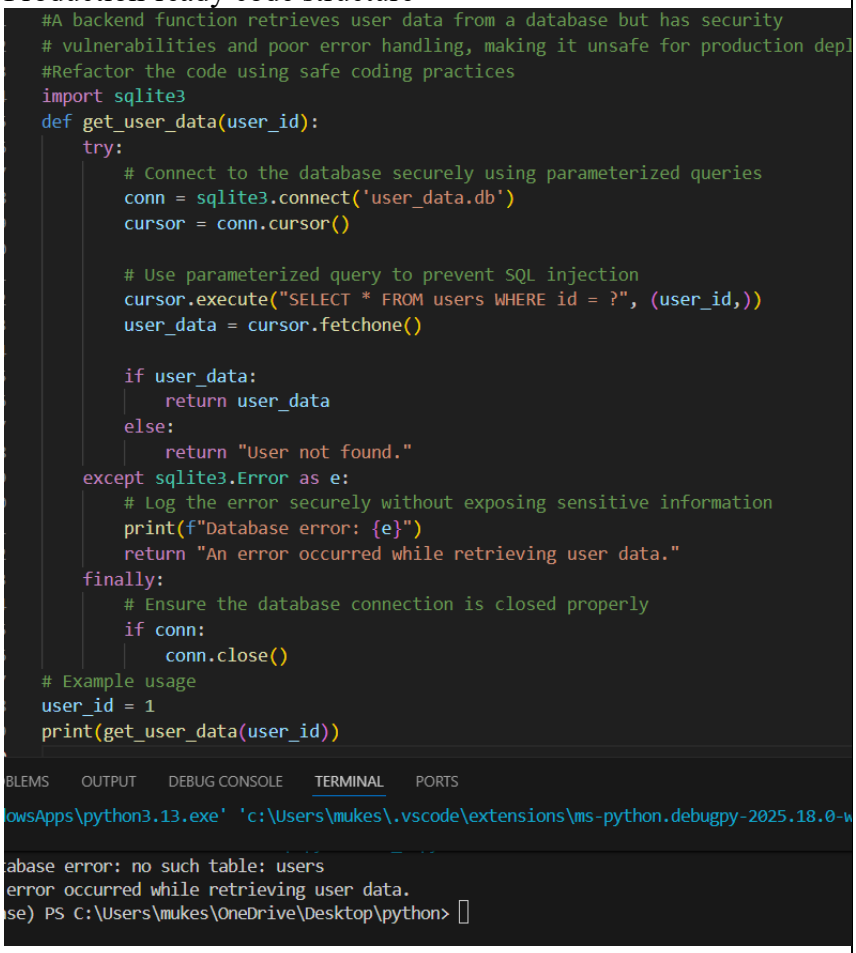
- Uses unsafe SQL query construction
- Has no input validation
- Lacks exception handling

Use AI tools to:

- Identify security vulnerabilities
- Refactor the code using safe coding practices
- Add proper exception handling
- Improve robustness and reliability

Expected Outcome

- Secure SQL queries using parameterized statements
- Input validation logic
- Try-except blocks for runtime safety
- AI-generated explanation of security improvements
- Production-ready code structure



```
#A backend function retrieves user data from a database but has security
# vulnerabilities and poor error handling, making it unsafe for production depl
#Refactor the code using safe coding practices
import sqlite3
def get_user_data(user_id):
    try:
        # Connect to the database securely using parameterized queries
        conn = sqlite3.connect('user_data.db')
        cursor = conn.cursor()

        # Use parameterized query to prevent SQL injection
        cursor.execute("SELECT * FROM users WHERE id = ?", (user_id,))
        user_data = cursor.fetchone()

        if user_data:
            return user_data
        else:
            return "User not found."
    except sqlite3.Error as e:
        # Log the error securely without exposing sensitive information
        print(f"Database error: {e}")
        return "An error occurred while retrieving user data."
    finally:
        # Ensure the database connection is closed properly
        if conn:
            conn.close()
# Example usage
user_id = 1
print(get_user_data(user_id))
```

```
BLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
owsApps\python3.13.exe' 'c:\Users\mukes\.vscode\extensions\ms-python.debugpy-2025.18.0-w
abase error: no such table: users
error occurred while retrieving user data.
se) PS C:\Users\mukes\OneDrive\Desktop\python>
```

Task 5: AI-Based Automated Code Review Report

Scenario

Your team uses AI tools to perform automated preliminary code reviews before human review, to improve code quality and consistency across projects.

Task Description

You are provided with a poorly written Python script.
Using AI-assisted review:

- Generate a **structured code review report** that evaluates:
  - Code readability
  - Naming conventions
  - Formatting and style consistency

- Error handling
- Documentation quality
- Maintainability

The task is not just to fix the code, but to **analyze and report on quality issues**.

Expected Outcome

- AI-generated review report including:
  - Identified quality issues
  - Risk areas
  - Code smell detection
  - Improvement suggestions
- Optional improved version of the code
- Demonstration of AI as a **code reviewer**, not just a code generator

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots